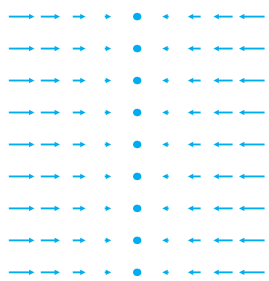
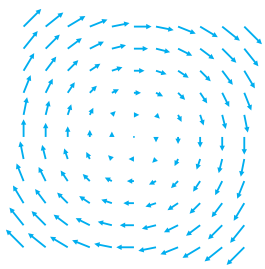
Suggested problems: 1-29 odd

Problem 1: For parts (a)-(f), does the vector field appear to be conservative?

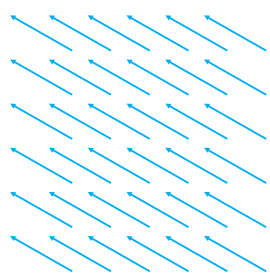
(a)



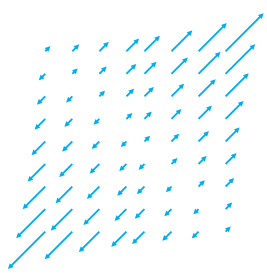
(b)



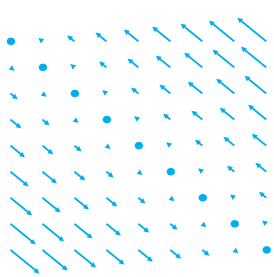
(c)



(d)



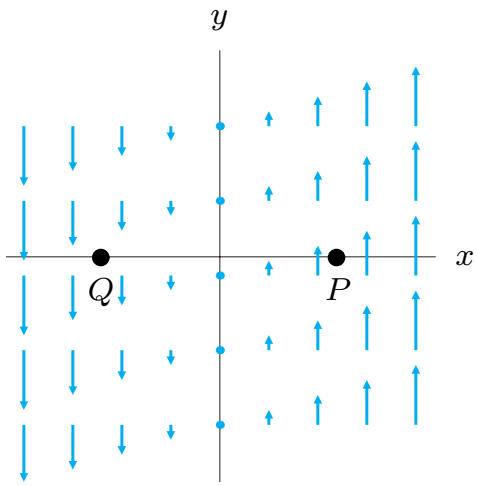
(e)



Problem 2: The figure below shows the vector field $\vec{F} = \langle 0, x \rangle$. Find paths C_1, C_2, and C_3 from P to Q such that

$$\int_{C_1} \vec{F} \cdot d\vec{r} = 0, \quad \int_{C_2} \vec{F} \cdot d\vec{r} > 0, \quad \int_{C_3} \vec{F} \cdot d\vec{r} < 0.$$

Is $\vec{F}$ a gradient field?



Problem 3: Prove that $\vec{F}(x, y) = \langle 2xy, xy \rangle$ is not conservative.

Problem 4: Show that the field

$$\vec{F}(x, y) = \frac{\langle -y, x \rangle}{x^2 + y^2}$$

has zero curl on its domain (the plane with the origin removed), but is *not* conservative.

Problem 5: Decide which of the following vector fields are conservative. For the ones that are conservative, find a potential function.

(a) $\vec{F}(x, y) = \langle y, -x \rangle$

(b) $\vec{F}(x, y) = \langle 2xy, x^2 \rangle$

(c) $\vec{F}(x, y) = \langle 2xy^3 + y, 3x^2y^2 + x \rangle$

(d) $\vec{F}(x, y) = \dfrac{\langle x, y \rangle}{x^2 + y^2}$